



CS & IT ENGINEERING

Discrete Mathematics



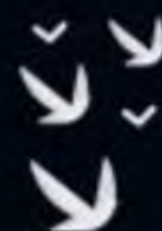
Graph Theory

Lecture No.- 16



By- Satish Sir

Recap of Previous Lecture



Topic

Planarity



Topics to be Covered

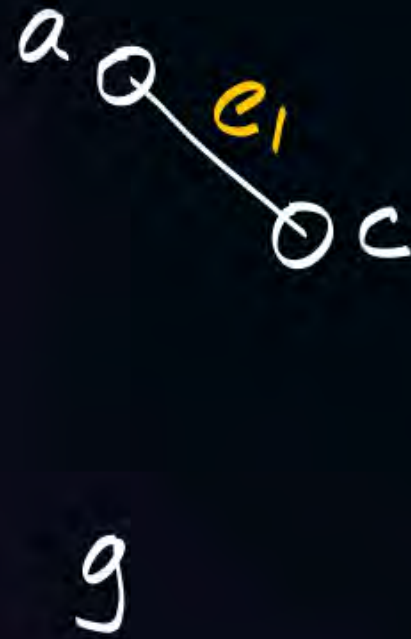
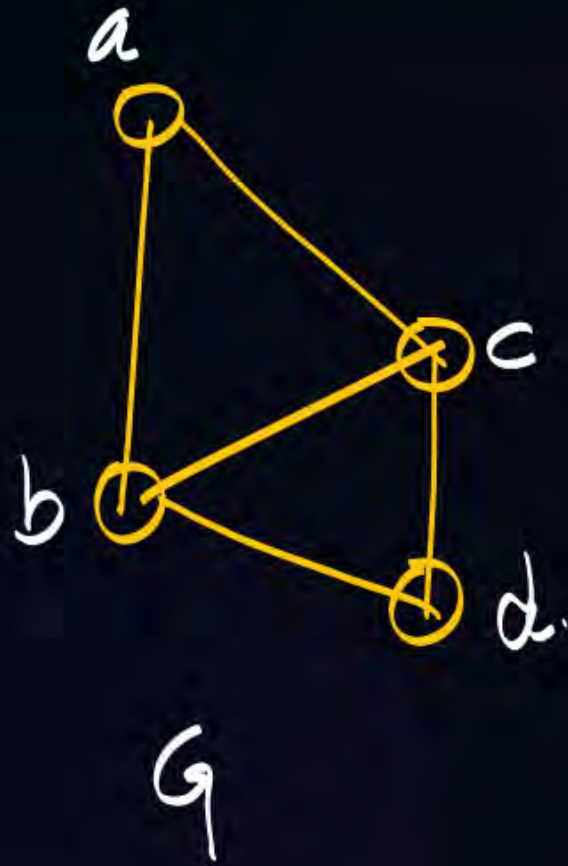


Topic

Extra Topics in Graph Theory



Subgraph :



vertices & edges of g must be in G .

each edge, it's end vertices if it present in g , then same way it should present in G .

Subgraph ($\subset$)

Subgraph of a Graph.
- is complete

$\Downarrow$
clique



G

$$g \subset G$$

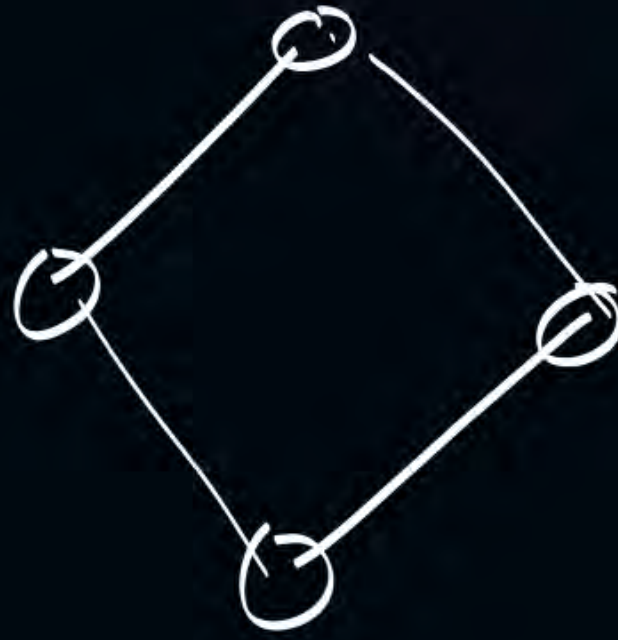
$$g' \subset G$$

$$g'' \subset G$$



g

$$G \subset G''$$



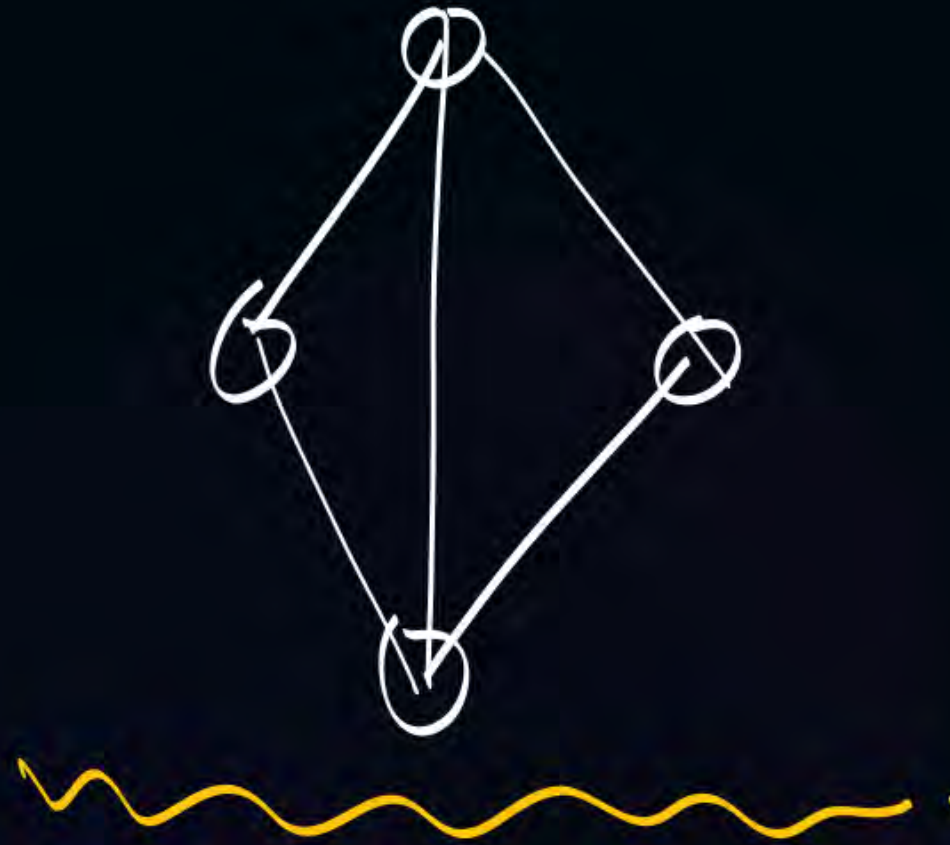
g'



g''



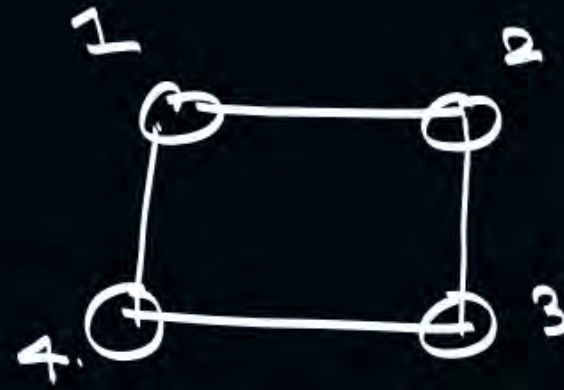
G_1



$\downarrow$
 k_1

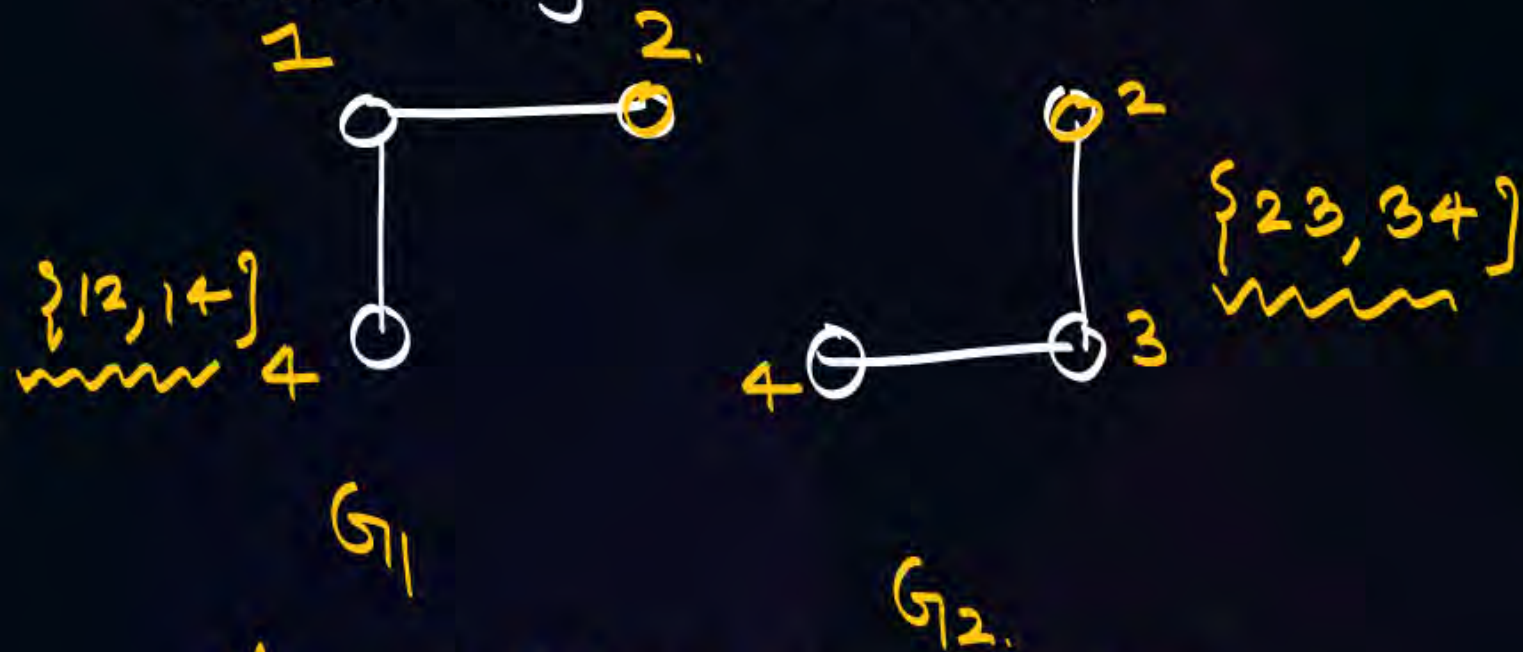
$\downarrow$
 k_2





edge-disjoint subgraph.

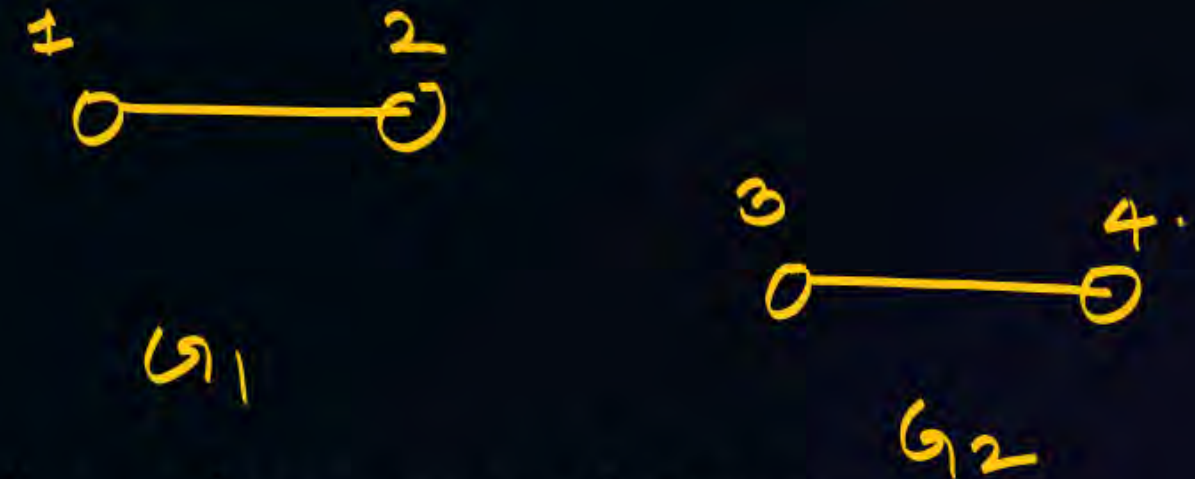
G_1, G_2 are subgraphs
such that they do not
have any common edges.



they might have common vertex.

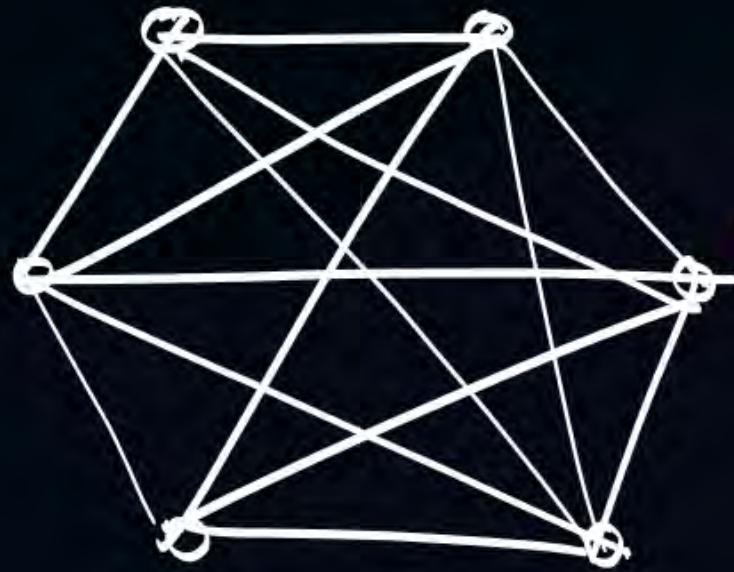
vertex disjoint subgraph.

G_1, G_2 are subgraphs
such that they do not
have any common vertex.



if the vertices are not common
no chances are that edges should be common.

⇒ How many subgraphs are possible in K_6 with 3 vertices.



→ we have to choose 3 vertices.

$$\downarrow$$

$6C_3$



what will be total graphs? with 3 vertices

$$2^{\frac{3(3-1)}{2}}$$

Total subgraphs with 3 vertices:

$${}^6C_3 \times 2^{\frac{3(3-1)}{2}}$$

⇒ what will be total
subgraphs with 3 vertices
in K_6 .

Choose 3 vertices in K_6 .

↓
 6C_3
~~~~~

Total Graphs with  $n$  vertices.

$$\frac{n(n-1)}{2}$$

2

Total graphs with 3 vertices.

$$\frac{3(3-1)}{2}$$

X

~~~~~

How many subgraphs are possible in K_6 ?

Subgraph with 1 vertex $\rightarrow {}^6C_1 \times 2^{\frac{1(1-1)}{2}}$

———))——— 2 vertices $\rightarrow {}^6C_2 \times 2^{\frac{2(2-1)}{2}}$

—————))————— 3 vertices $\rightarrow {}^6C_3 \times 2^{\frac{3(3-1)}{2}}$

$\rightarrow {}^6C_4 \times 2^{\frac{4(4-1)}{2}}$

$\rightarrow {}^6C_6 \times 2^{\frac{6(6-1)}{2}}$

$$\sum_{i=1}^6 {}^6C_i \times 2^{\frac{i(i-1)}{2}}$$

Total subgraphs in K_n ?

$$\sum_{i=1}^n {}^nC_i \times 2^{\frac{i(i-1)}{2}}$$

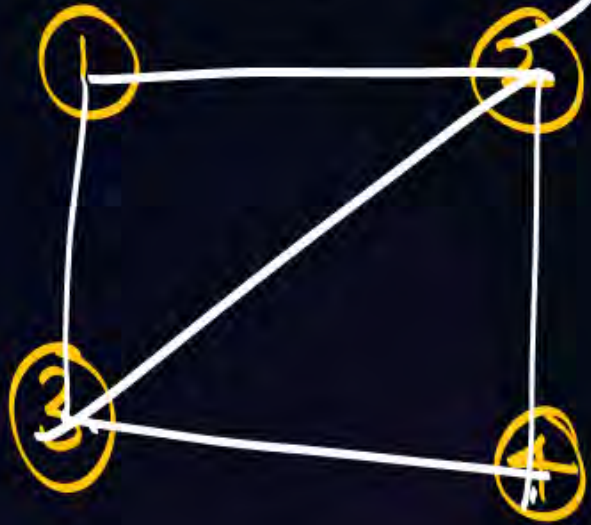
Graph operations:

Union ($\cup$)

Intersection ($\cap$)

Ring sum ($\oplus$)

$$G_3 = (V_3, E_3)$$



$$G_1 = (V_1, E_1)$$

$$V_1 = \{1, 2, 3\}$$

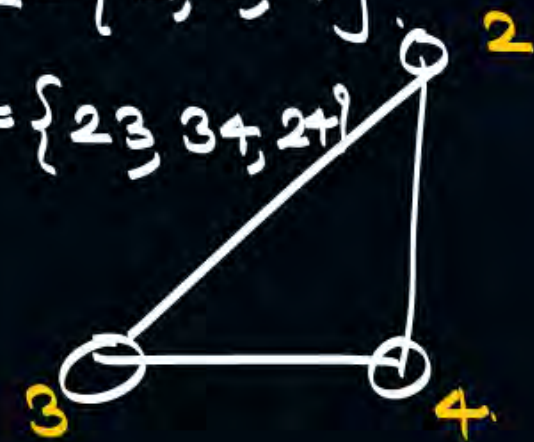
$$E_1 = \{12, 13\}$$



$$G_2 = (V_2, E_2)$$

$$V_2 = \{2, 3, 4\}$$

$$E_2 = \{23, 34, 24\}$$



union: $G_3 = (V_3, E_3)$

$$\begin{aligned} V_3 &= V_1 \cup V_2 = \{1, 2, 3\} \cup \{2, 3, 4\} \\ &= \{1, 2, 3, 4\} \end{aligned}$$

$$G_3 = G_1 \cup G_2$$

$$V_3 = V_1 \cup V_2$$

$$E_3 = E_1 \cup E_2$$

$$G_3 = G_1 \cap G_2.$$

$$G_3 = (V_3, E_3)$$

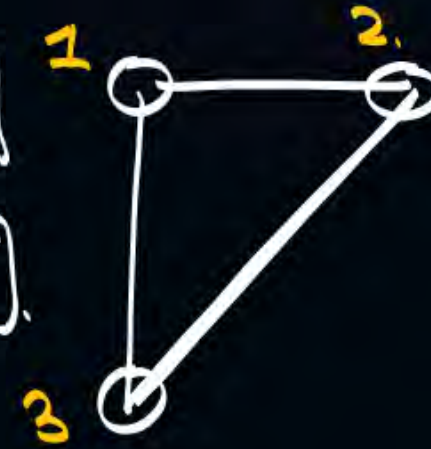
$$V_3 = V_1 \cap V_2.$$

$$E_3 = E_1 \cap E_2.$$

$$G_1 = (V_1, E_1)$$

$$V_1 = \{1, 2, 3\}$$

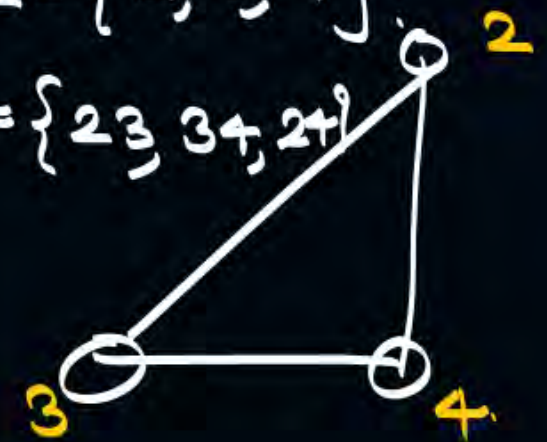
$$E_1 = \{12, 13\}$$



$$G_2 = (V_2, E_2)$$

$$V_2 = \{2, 3, 4\}$$

$$E_2 = \{23, 34, 24\}$$



=



Ring sum ($\oplus$)

$$G_3 = (V_3, E_3)$$

$$V_3 = \underline{V_1 \cup V_2}$$

$$E_3 = \underline{(E_1 \cup E_2) - (E_1 \cap E_2)} \quad \left\{ \text{edges either in } G_1 \text{ or in } G_2 \text{ but not in both} \right\}$$

$$G_1 = (V_1, E_1)$$

$$V_1 = \{1, 2, 3\}$$

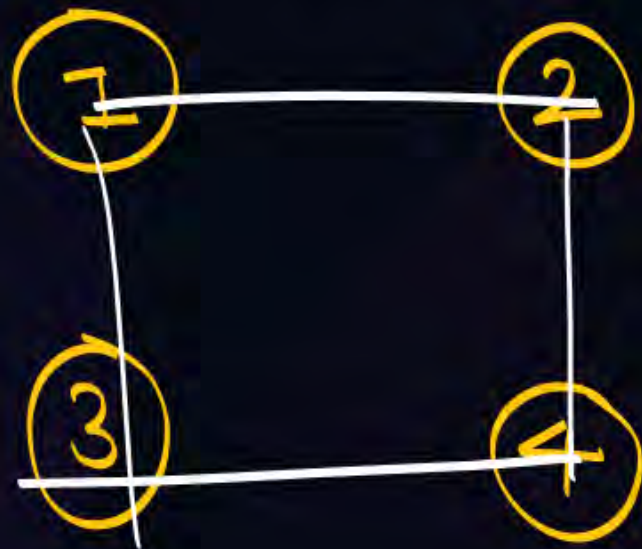
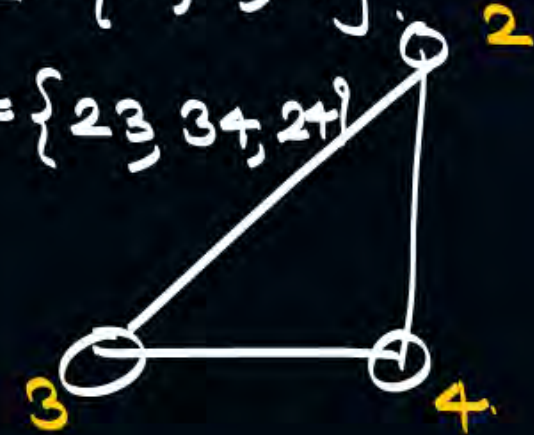
$$E_1 = \{12, 13\}$$



$$G_2 = (V_2, E_2)$$

$$V_2 = \{2, 3, 4\}$$

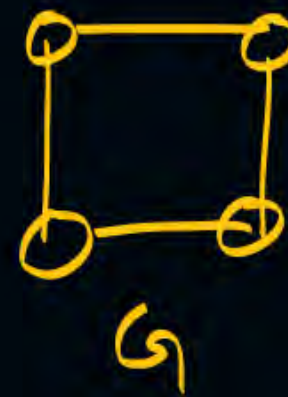
$$E_2 = \{23, 34, 24\}$$



$$\left\{ \begin{array}{l} G_1 \cup G_2 = G_2 \cup G_1 \\ G_1 \cap G_2 = G_2 \cap G_1 \\ G_1 \oplus G_2 = G_2 \oplus G_1 \end{array} \right.$$

$$G \cup G = G \cap G = G.$$

$$G \oplus G = \text{null Graph.}$$



- * single vertex of a graph is Subgraph of a Graph.
- * single edge of a Graph is subgraph of a Graph.
- * $a \subset b \subset G$, $a \subset G$.
- * Subgraph of a subgraph of a Graph is subgraph of a Graph.



2 mins Summary

Topic

One

Topic

Two

Topic

Three

Topic

Four

Topic

Five

Extra Topics in Graph Theory



THANK - YOU